Session 2.1: The Data Feedback Loop Strategy



THE SITUATION

StreamWise: members cancel, and Jun does not know why

StreamWise is a new streaming service for university students. Members pay $5.00 a month to watch shows and movies. Jun is its founder.

MEASURE (ONE MONTH)	VALUE
Members at the start of the month	**12,000**
Plays started	**48,000**
Plays finished	**9,600**
Titles saved to "My List"	**3,100**
Members who cancelled	**420**
Monthly price per member	**$5.00**

Jun's three questions

- Which signals should he use?
- What should he change because of them?
- Should he sell viewer profiles to an advertiser, or would that hurt students' trust?

What is the Data Feedback Loop Strategy?

A company decides **which customer signals to collect**, **which business decision each signal will improve**, and **how to keep customers' trust**. Collecting more data does not make a company stronger by itself. A company gains an advantage when a small set of useful signals changes a decision, the company measures the result, and the next decision becomes better. This repeating cycle is the **data feedback loop**.

Use it when a business can see many customer actions but does not know which ones show a real need, or when it is tempted to collect everything "just in case."

Signals: three questions to ask about every signal

Signal

One recorded customer action.

Example: a viewer presses "Save" on a show.

Q1. How clear is it?

Strong signal: one clear meaning, because the customer spent time or effort.

Weak signal: can mean many things.

Example: finishing a 40-minute episode is strong; opening a title for 3 seconds is weak (maybe a mistaken click).

Signals: what they tell us, and when they arrive

Q2. What does it tell us?

Like: what customers enjoy, so we offer more of it.

Unhappy: what is wrong, so we fix it.

Example: saving a show shows liking; cancelling shows the member is unhappy.

Q3. Early or late?

Early: arrives while we can still act.

Late: arrives after the customer is already gone.

Example: finishing fewer shows each week is early; cancelling is late.

QUICK PRACTICE / 5 MIN

Read the StreamWise signals

Write one answer from each column heading. If a number is not a signal (not a customer action), skip Q1 to Q3.

STREAMWISE DATA	SIGNAL? YES OR NO	Q1: HOW CLEAR? STRONG OR WEAK	Q2: TELLS US? LIKE OR UNHAPPY	Q3: WHEN? EARLY OR LATE
Members at the start: 12,000	_____	_____	_____	_____
Plays started: 48,000	_____	_____	_____	_____
Plays finished: 9,600	_____	_____	_____	_____
Titles saved to "My List": 3,100	_____	_____	_____	_____
Members who cancelled: 420	_____	_____	_____	_____
Monthly price: $5.00	_____	_____	_____	_____

Cancellations are strong, but they come too late

STREAMWISE DATA	SIGNAL? YES OR NO	Q1: HOW CLEAR? STRONG OR WEAK	Q2: TELLS US? LIKE OR UNHAPPY	Q3: WHEN? EARLY OR LATE
Members at the start: 12,000	No	-	-	-
Plays started: 48,000	Yes	Weak	Like	Early
Plays finished: 9,600	Yes	Strong	Like	Early
Titles saved to "My List": 3,100	Yes	Strong	Like	Early
Members who cancelled: 420	Yes	Strong	Unhappy	**Late**
Monthly price: $5.00	No	-	-	-

Not signals: members is a count, and the price is set by StreamWise. Plays started shows only weak liking: just 20% were finished.

Cancellations are strong but late: they come too late to keep those members and do not say why. Watch early signals, such as fewer finished plays.

Churn rate, ARPU, and consent

Churn rate

The percentage of members who cancel in a period.

Churn rate = members who cancelled / members at the start x 100

Example: 420 / 12,000 x 100 = 3.5% per month.

ARPU

Average Revenue Per User

The average money earned from one member in a period.

Example: StreamWise charges $5.00 per month, so ARPU is $5.00 per member per month.

Consent

A customer's clear permission to collect and use their data for a stated purpose.

Example: "We use what you finish and save to recommend shows."

Five steps that repeat

01
Collect
Record a signal.

02
Interpret
Ask what it may mean.

03
Act
Change one action.

04
Measure
Compare one number.

05
Learn
Keep or change the action.

The loop is complete only when the result of one decision makes the next decision better.

The strategy has four rules. Each rule protects one part of this loop.

Rule 1: Start with the decision, not the data

First name the decision you want to improve. Then collect only the signals that can change it.

WHAT IT LOOKS LIKE

Example

A campus cafe app wants to decide "which drink to promote on rainy days." It needs two signals: the weather and the drinks ordered that day. It does not need contact lists or browsing history.

WHAT TO AVOID

Common mistake

"Let's collect everything and find something later." Data with no decision costs money to store and adds privacy risk, but improves nothing.

Rule 2: Use strong signals; check weak signals before you act

Prefer actions that show effort or repeat behavior: finishing, saving, buying again, searching for a specific item.

WHAT IT LOOKS LIKE

Example

At StreamWise, 48,000 plays were started but only 9,600 were finished: 9,600 / 48,000 x 100 = 20%. A play started is a weak signal of interest; a play finished is stronger.

WHAT TO AVOID

Common mistake

Treating a weak signal as proof. A viewer who stops after 10 seconds may dislike the show, or may have received a phone call.

Rule 3: Close the loop and measure one result

Collect, interpret, change one action, measure the result, and learn for the next decision.

WHAT IT LOOKS LIKE

Example

Students who finish a cooking show often save other cooking shows. StreamWise adds a "More easy cooking" row for them and measures their churn for one month. If churn falls from 3.5% to 2.5%, the row is kept; if not, it is changed.

WHAT TO AVOID

Common mistake

Changing something but never measuring the result. Without the measure step, the company cannot learn whether the data helped.

Rule 4: Earn trust: ask permission, collect only what you need, give value back

Customers share data more willingly when they understand the purpose and receive something useful in return.

WHAT IT LOOKS LIKE

Example

A sign-up screen says: "We use what you finish and save to recommend shows. We never sell your viewing history." Members can turn this off at any time.

WHAT TO AVOID

Common mistake

Collecting or selling data in ways customers would not expect. Even accurate data can damage the brand if customers feel watched.

Netflix recommendations

What happened

Netflix uses what members watch, finish, and search for to decide which titles and which artwork to show each member. It tests a change on a group of members and keeps it only if viewing improves. Netflix researchers reported that recommendations influence about 80% of the hours members stream (Gomez-Uribe and Hunt, 2015).

Why it is good

- It starts with a clear decision: which titles to show on each home screen. **(Rule 1)**
- It relies on strong signals such as finished titles, not only clicks. **(Rule 2)**
- It tests each change and measures the result before keeping it. **(Rule 3)**

GOOD EXAMPLE 2

Spotify Discover Weekly

What happened

Since 2015, Spotify has given each listener a new personalized playlist every Monday. It is built from songs the listener plays, saves, adds to playlists, and skips.

Why it is good

- Saves and repeated plays are strong signals; quick skips show what to avoid. **(Rule 2)**
- The listener receives clear value from the data: new music chosen for them. **(Rule 4)**
- Each week, the listener's reaction becomes new data for the next playlist. **(Rule 3)**

BAD EXAMPLE 1

Google Flu Trends

What happened

From 2008, Google estimated how many people had the flu by counting flu-related searches. In February 2013, it estimated more than twice the share of flu doctor visits that the U.S. Centers for Disease Control and Prevention (CDC) later reported (Lazer and others, 2014). Google stopped publishing the estimates in 2015.

Why it is bad

- A search is a weak signal: people search because of news or curiosity, not only because they are sick. **(Rule 2)**
- The model was not regularly checked against real doctor-visit data and corrected. **(Rule 3)**
- Big data volume created confidence without accuracy. Many signals are not the same as useful signals.

BAD EXAMPLE 2

Target's pregnancy prediction

What happened

The New York Times reported in 2012 that Target used shopping signals, such as unscented lotion and certain vitamins, to predict which customers were pregnant and mailed them baby-product coupons. A father complained after his teenage daughter received these coupons.

Why it is bad

- The signals were strong and the prediction was often accurate, but customers did not expect a store to know this. **(Rule 4)**
- Customers had not agreed to have sensitive facts guessed from their purchases: no clear consent.
- Lesson: accurate data can still damage trust and the brand.

Which rule made the difference?

EXAMPLE	RESULT	THE RULE THAT DECIDED IT
Netflix	**Good**	Tests each change and measures it (Rule 3)
Spotify	**Good**	Strong signals, and value back to the listener (Rules 2 and 4)
Google Flu Trends	**Bad**	Weak signal, never corrected (Rules 2 and 3)
Target	**Bad**	Accurate, but no consent or trust (Rule 4)

Good data use is not about the amount of data. It is about the right signal, a measured result, and trust.

Tool 1: Test group and comparison group

Try a change on one group, not on a similar group, and compare the same number.

WHAT IT LOOKS LIKE

Example: a cafe coupon

- 500 members get a coupon (test group). 500 members get none (comparison group).
- Coupon group: 75 order this week, or 15 of every 100.
- Other group: 50 order, or 10 of every 100.
- The coupon added about 5 orders for every 100 members.

USED IN EXERCISE 2

Why it matters

Without the comparison group, you cannot tell whether orders rose because of the coupon or because of something else, such as rainy weather.

Tool 2: From churn to money

Lower churn means more members stay and keep paying.

USED IN EXERCISE 3

Two formulas

- Members kept per month = members x (old churn rate - new churn rate)
- Revenue kept per month = members kept x monthly price

WHAT IT LOOKS LIKE

Example: a cafe subscription

- 1,000 members pay $10.00 a month.
- Churn falls from 6% to 4%: 1,000 x 2% = 20 more members stay.
- Revenue kept: 20 x $10.00 = $200.00 that month.
- The gain grows every month, because those members keep paying.

Tool 3: Trade-off and decision rule

MEMO BULLET 2

Trade-off

What you give up to get something you value more.

Example: a fitness app refuses $2.00 per user from an advertiser (gives up money) to protect users' trust and its subscriptions (gains).

MEMO BULLET 3

Decision rule

A promise, written before you start, about which result would make you change your decision.

Example: "If churn does not fall below 4% in 90 days, we will try a different plan."

The result, not opinion, decides.

Choose the strongest signal: CampusEats

Decision: which restaurants to show at the top of the app at lunchtime.

Signals the app records

- (a) A student opens a restaurant page for 2 seconds.
- (b) A student orders from the same restaurant three times in one month.
- (c) A student gives a restaurant 5 stars after an order.
- (d) A student scrolls past a restaurant.
- (e) A student adds a restaurant to "Favorites."

EXERCISE 1 / EASY / 20 MIN

Use the template

TEMPLATE

Decision: [the decision the business wants to improve]
Strongest signal: [letter and action], because [what effort or repeat behavior it shows].
Weakest signal: [letter and action], because [two different things it could mean].

WORKED EXAMPLE: MUSIC APP

Decision: Which songs to put in the "Study Focus" playlist.
Strongest signal: a student plays the same song five times during study hours, because repeating a song takes effort and shows a clear preference.
Weakest signal: a student opens a song for 2 seconds, because it could mean the student disliked it or simply clicked by mistake.

Repeat orders are the strongest signal

STRONG SIGNALS

Strongest

(b) Three orders in one month. The student paid money again and again.

(e) "Favorites" and (c) 5 stars are also useful, but show less effort than repeat orders.

WEAK SIGNALS

Weakest

(a) a 2-second view and **(d) a scroll**.

They can mean dislike, a mistake, or simply that the student was not hungry.

Build a feedback loop: FitU gym app

Scenario

FitU is a campus gym app with **2,000 members**. Last month **160 members cancelled**.

Many members **book** a fitness class but do **not attend** it.

Your steps

- **Step 1:** Churn rate = 160 / 2,000 x 100 = ____ %
- **Step 2:** Complete the five-step loop (Rule 3). Use a test group and a comparison group (Tool 1).

Use the template

TEMPLATE

1. Collect: [the signal we record]
2. Interpret: [what the signal may mean, and one other possible meaning]
3. Act: [one change we will test on a test group, and who is the comparison group]
4. Measure: [one number we will compare before and after, and our target]
5. Learn: If [the result], we will [keep / change] the action.

WORKED EXAMPLE: CAMPUS EVENTS APP

1. Collect: students who save an event but do not check in at the door.
2. Interpret: they may forget the event, or they may save events they never planned to attend.
3. Act: send a reminder 2 hours before saved events to half of these students; the other half gets no reminder (comparison group).
4. Measure: attendance rate of saved events; target: from 20% to 30%.
5. Learn: If attendance rises to 30% only in the reminded group, we will keep reminders for everyone.

Churn is 8%. Does your loop have all five steps?

Churn rate = 160 / 2,000 x 100 = 8%

A GOOD LOOP

- **Collect:** records "booked but did not attend."
- **Interpret:** gives two meanings, for example the class time is inconvenient, or the member forgot.
- **Act:** tests one change on a test group and compares it with a comparison group, for example a reminder for half of these members.
- **Measure:** one number with a target, for example churn from 8% to 6%.
- **Learn:** says what happens if the target is or is not reached.

Choose a data strategy for StreamWise

12,000 members / $5.00 per month / churn 3.5%

OPTION A

Option A: Sell viewer profiles

An advertiser will pay **$1.00 per member per month** for detailed viewing histories.

In a StreamWise survey, **30% of members** said they would cancel if their viewing history were sold.

OPTION B

Option B: Permission-based recommendations

Ask permission to use three signals (finished titles, saved titles, searches) only to improve recommendations. Never sell them.

Goal: lower monthly churn from **3.5% to 2.5%**.

Calculate before you decide (Tool 2)

- Option A extra revenue per month = 12,000 x $1.00 = $_____
- If 30% of members cancel: members lost = 12,000 x 30% = _____
- Revenue lost per month = members lost x $6.00 ($5.00 subscription + $1.00 advertiser payment) = $_____
- Option B members kept per month = 12,000 x (3.5% - 2.5%) = _____
- Option B revenue kept per month = members kept x $5.00 = $_____

Write a 3-bullet memo to the CEO (Tool 3)

TEMPLATE

To: [CEO name], [company]
1. Decision and evidence: We should choose [option] because [the rule it follows] and [one number from your calculation].
2. Biggest risk and trade-off: The weakest point is [risk]. We give up [what we lose] to gain [what we protect or win].
3. Next step and decision rule: In the next 90 days we will [one action] and measure [one number]. We will change this decision if [result].

Then give a 2-minute presentation using your three bullets.

FitTrack: an insurer wants users' location history

FitTrack is a running app. An insurance company offers $2.00 per user per month for location history.

To: Mia Park, CEO, FitTrack
1. Decision and evidence: We should refuse to sell location history and instead ask permission to use running data only for training plans, because users expect privacy (Rule 4) and 25% of surveyed users said they would leave if we sold it.
2. Biggest risk and trade-off: The weakest point is that we do not yet know whether better training plans will keep users. We give up $2.00 per user per month in insurance revenue to protect our users' trust and our subscription revenue.
3. Next step and decision rule: In the next 90 days we will offer personalized plans to users who give permission and measure their monthly churn. We will change this decision if churn does not fall below 4%.

Selling the data costs more than it earns

The numbers

- Option A brings **$12,000** per month.
- If 3,600 members (30%) cancel, StreamWise loses 3,600 x $6.00 = **$21,600** per month.
- Option B keeps 12,000 x 1% = **120 members**, or **$600** per month, and this grows every month churn stays lower.

A strong memo

- Usually chooses Option B, or a version of it.
- Uses at least one number from Step 1.
- Names a real risk: members may refuse permission, or recommendations may not lower churn.
- States a clear rule for changing the decision.
- Option A can still be strong if it explains real consent and gives a decision rule.

SUCCESS CHECK

Can you do these three things?

01

Name one strong and one weak signal for a business decision and explain the difference.

02

Complete all five steps of a feedback loop with one measurable result.

03

Explain why selling customer data can cost more than it earns.

REFERENCES

Gomez-Uribe, C. A., and Hunt, N. (2015). The Netflix Recommender System: Algorithms, Business Value, and Innovation. ACM Transactions on Management Information Systems.

Lazer, D., Kennedy, R., King, G., and Vespignani, A. (2014). The Parable of Google Flu: Traps in Big Data Analysis. Science.

Duhigg, C. (2012). How Companies Learn Your Secrets. The New York Times Magazine.

Hagiu, A., and Wright, J. (2020). When Data Creates Competitive Advantage. Harvard Business Review.